Supplementary material to "Choupo: an open-source chemical process simulator for education and research"

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Revision 2 – DRAFT

S1 Reproducing every number

All numbers in the article and in this supplement were produced on one commit of the development line of https://github.com/choupo-admin/choupo (the commit carrying this manuscript; based on `c11d0354a`). With a C++17 compiler and GNU make, from the repository root:

```
make all -j3                          # the five binaries, no external library
./choupoSolve  <case>                 # a steady case (application choupoSolve)
./choupoProps  <case>                 # a property / fitting case (choupoProps)
python3 -I benchmarks/dwsim/compare.py all     # the five benchmark cases
python3 -I paper/cace/figures/gen_dspmde_fit.py    # data of Figure 4a
python3 -I paper/cace/figures/gen_sdem_traces.py   # data of Figure 4b
python3 -I bin/curate/check_mass_closure.py        # Table 3, mass column
python3 -I bin/curate/check_energy_closure.py      # Table 3, energy column
```

A case writes into its own directory, so each was copied to a scratch directory before it was run. Table S1 maps each item of the article to its case and to the file it was read from.

Table S1: Where each item of the article comes from.

item	case	read from
Listings 1–3	`tutorials/steady/flash/flash01_benzene_toluene`	the run log; the refusal: the same case with `vaporFraction0.3;` appended to `0/feed`
heater check (Sect. 3.2)	`tutorials/steady/heat/hx01_heater`	`converged/hotStream`; `reports/balances/globalEnergyBoundary.csv`
Figure 2	`tutorials/props/curation/curate05_nrtl_acetone_methanol`	the CSV files the run writes (identical to `paper/whitepaper/data/`); χ^2 from `heldout_residuals.csv` and the held-out dataset's `U95`
Figure 3, Table 4	`tutorials/steady/flowsheets/cavett01_recycle_train`	the log (Newton history); `converged/P1`; `reports/balances/*.csv`; Wegstein: `recycleSolverWegstein;` in `system/solverDict`
Figure 4a	`tutorials/steady/membranes/membrane24_fit_then_simulate` (props mode)	`postProcessing/fit_NF270_FdL2021/rejections.csv`; the fit's printed parameters and verdict
Figure 4b	probes built from `tutorials/steady/membranes/membrane12_sdem_nf270_nacl_traces` by `bin/curate/check_sdem.py`'s builder	`paper/cace/data/sdem_*.csv`
NanoFiltran comparison	`tutorials/props/membrane/nanofiltran_geraldesAlves2008`	`bin/curate/check_nanofiltran_reference.py`
electrodialysis	`tutorials/steady/electrodialysis/ed04_limiting_current_multiionic`	the printed `i_lim` of each unit; the measured values: the case's README (Table 4 of the source)
Tables 5–7	`benchmarks/dwsim/<case>`	`compare.py`'s tables; each case's README

S2 The reference's stored states: what an audit of the sample files found

The comparison of the article is with the solved states stored in five sample files of DWSIM, read at commit `1abf72d1b6b41d3e9a8cc770d3cc4e8fc76e5766` of https://github.com/DanWBR/dwsim (2026-07-17); the files record build versions 8.8.0.0 (A, B, C), 6.5.2.0 (D) and 5.3.6858.20432 (E). The samples were not re-executed. The findings below are facts about those files, measured on their own numbers; they bound how finely each comparison can resolve, and each is being offered to the program's maintainers. None of them is evidence about the quality of Choupo.

A (Cavett's problem) The stored feed has 15 compounds and no *n*-undecane, and its total (5908.21 mol/s) matches neither reading of the primary specification's units; the primary's absolute pressures were entered as gauge pressures (800/270/49/13 + 14.7 psi); each of the three recycle blocks reports convergence after one iteration, and one passes on a state that differs from its source's (vapour fraction 0.38562 against 0.38906, 20.39 kW apart); one separator stores its vapour and its liquid 0.97 K apart; the stored state closes its first law to +17.49 kW.

B (pressure-swing distillation) The two bottoms specifications admit no steady state (+0.14 % on the plant); the recycle stream carries 1.47 % more than the stream it copies, while its recycle block reports convergence after one iteration; the light distillate is stored 1.98 K below its own condenser stage; the property package lists the acetone–methanol NRTL interaction twice with different values.

C (extraction with MIBK) The column is adiabatic and cools by 13.4 K (stages at 296.54, 295.81, 290.79 and 284.70 K); its first law on the stored enthalpies closes to −0.54 kW; two stored extractor specifications do not match their stored calculated values (in that mode they are not used); the package is a modified (Dortmund) UNIFAC.

D (methanol from CO_2) A 98.9 % methanol distillate stored at 184.69 K; two exchanger efficiencies stored above 100 %; one adsorption constant carried as 3453.38 and 3454.38 in two rate expressions, and the kinetics depart from the article in three more places (an exponent of 40 000/RT for 36 696, a rate constant at the lower bound of its interval, a parenthesis that moves a square root); a compressor with negative work; a stored plant flow balance of −17.0 %; the hydrogen feed is stored two-phase at the compressor's suction.

E (steam reforming) The rates read partial pressures in atm where the article's are in bar; two of the three rates truncate the article's shared adsorption denominator; the third equilibrium constant is not the product of the first two (19.65 against 40.54 at 1000 K, bar basis); the stored duty, 1221.81 kW, is 2.6 times the heat of reaction of the stored conversion; the pre-exponential factors are rounded.

S3 Case C: acetone extraction with MIBK

Choupo's predictive UNIFAC did not describe the water–MIBK miscibility gap well enough (its water-rich branch sits at the flash's composition floor, against 0.0038 measured), so a liquid–liquid NRTL set was curated from the measured tie lines of Choi and Rhim (1986, J. Korean Inst. Chem. Eng. 24, 185–194), converted from the article's table by arithmetic only. At the midpoints of the authors' calculated tie lines it returns those lines to within 0.0035 mole fraction on lines 1–9 and 0.018 on line 10, nearest the plait point; against the measured lines it misses the binary edge (water in MIBK 0.013 against 0.118 measured), as the authors' own calculated line does. On the tree of this revision Choupo's four-stage extractor and the same cascade written as four mixers and four flashes agree to every printed digit: extract 984.05 mol/s, raffinate 293.29 mol/s, acetone recovery 1.0000, against the reference's stored 911.32, 366.01 and 0.9991. The UNIFAC variant refuses by name. The two programs solve different problems (adiabatic against isothermal), and temperature moves the split, so this case is not used in the article's comparison.

S4 Case E: the grid study

Table 5 of the article gives Choupo's methane conversion against the number of RK4 steps through the bed. The case was first built with 400 steps; at that resolution its outlet was 0.85 % short of its own limit, and the case now uses 25 600 (0.4 s of computing). Twenty steps – the sample's stored volume increment – refuse by name in Choupo: the first step drives the hydrogen partial pressure to zero and the rate law has no value there. The other benchmark with a bed, case D, was checked the same way: its methanol product moves by 10^{-9} relative between 5000 and 20 000 steps.

S5 The browser front-end

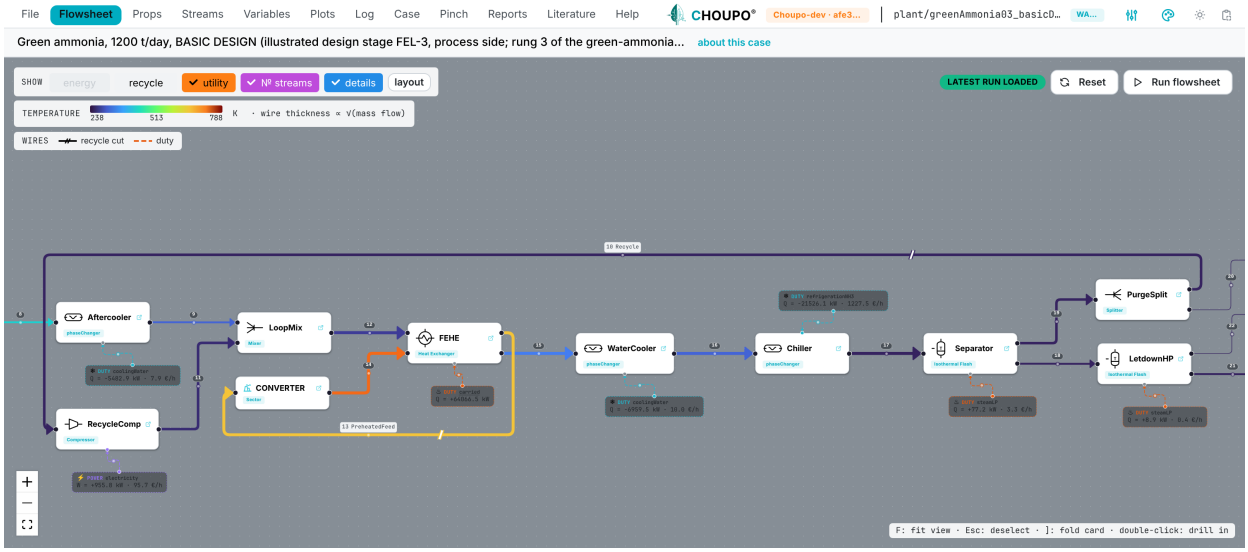


Figure S1: The browser front-end on `tutorials/plant/greenAmmonia03_basicDesign`, run in the page by the WebAssembly build of the engine (captured 2026-10-21 by `paper/whitepaper/figures/capture_gui.mjs`). Streams are coloured by the temperature the engine published; slashed wires are the recycle tears; `CONVERTER` is a sector of three catalyst beds and two inter-bed coolers. Nothing on the screen is computed in the browser.

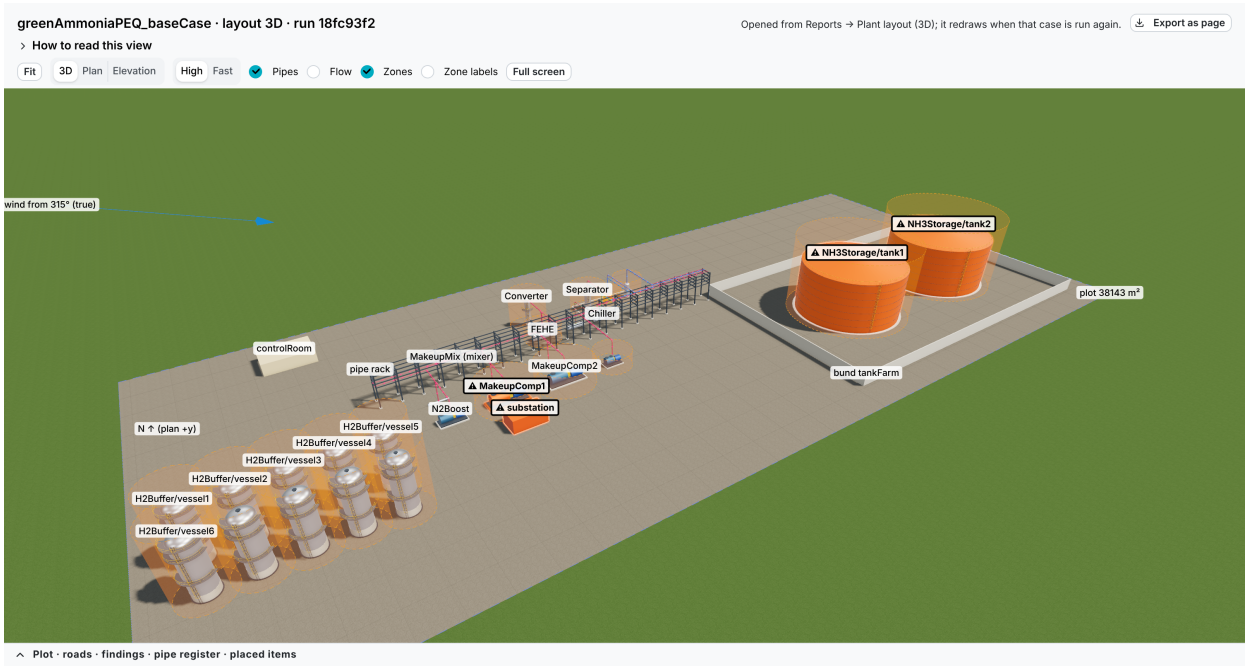


Figure S2: The plot plan of `tutorials/plant/greenAmmoniaPEQ_baseCase`, drawn from the layout block of the run's result: each item where the case places it, on the envelope the sizing pass published, with the case's declared hazardous-area zones and the pipes the engine routed.